Quiz: Basic arithmetic

This quiz has 5 questions. The passing score is 4.

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Started on	Tuesday, January 14, 2025, 3:50 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 3:52 PM
Time taken	2 mins 20 secs
Grade	**0.00** out of 5.00 (**0**%)

Question **1**

Incorrect

0.00 points out of 1.00

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The decimal number 32768 can be represented as (pick all that apply):

Select one or more:

- [x] A. Fullword 00008000 ✔
- [ ] B. Fullword FFFF8000
- [ ] C. Signed halfword 8000
- [ ] D. Doubleword 00000000FFFF8000
- [ ] E. Doubleword 0000000000008000

Decimal 32768 to hexadecimal:

32768 / 16 = 2048 , with remainder **0**

2048 / 16 = 128 , with remainder **0**

128 / 16 = 8 , with remainder **0**

8 / 16 = 0 , with remainder **8**

Decimal 32768 is **hexadecimal 8000**.

This can be represented as **fullword 00008000**, or **doubleword 0000000000008000**.

Note that decimal 32768 *cannot be represented as **signed halfword 8000*** (binary **1**000 0000 0000 0000) because *that is a negative number:* **-32768**.

The correct answers are: Fullword 00008000, Doubleword 0000000000008000

Question **2**

Incorrect

0.00 points out of 1.00

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The decimal number **-**32768 can be represented as (pick all that apply):

Select one or more:

- [x] A. Doubleword FFFFFFFFFFFF8000 ✔
- [ ] B. Doubleword 0000000000008000
- [ ] C. Fullword FFFF8000
- [ ] D. Fullword 00008000
- [ ] E. Halfword 8000

-32768 is, in a binary halfword, **8000**. Bit 0, the sign bit, is 1 (hexadecimal **8** is binary **1**000), indicating a negative number.

To represent the number as a **fullword**, we propagate the sign (**binary 1**) to the left: **FFFF8000**.

Same to represent the number as a **doubleword**: **FFFFFFFFFFFF8000**.

The correct answers are: Halfword 8000, Doubleword FFFFFFFFFFFF8000, Fullword FFFF8000

Question **3**

Incorrect

0.00 points out of 1.00

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The decimal number 4096, converted to hexadecimal and placed in a halfword, is:

Select one:

- A. 4096
- B. F100 ✖
- C. The result is too large for a halfword and must be placed in a fullword
- D. 1000
- E. FFFF

4096 / 16 = 256 with remainder **0**

256 / 16 = 16 with remainder **0**

16 / 16 = 1 with remainder **0**

1 / 16 = 0 with remainder **1**

Decimal 4096 is **hexadecimal halfword 1000.**

The correct answer is: 1000

Question **4**

Incorrect

0.00 points out of 1.00

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The signed halfword A000 can be represented in a fullword as 0000A000

Select one:

- (selected) True ✖
- False

To convert a halfword to fullword, we propagate the sign to the left.

In the halfword A000, bit 0, the sign, is '1', which means A000 is negative (hexadecimal 'A' is binary '**1**010'). *We must propagate '**1**' to the left.*

So, the correct representation of 'A000' as a fullword is 'FFFFA000' ('F' is binary '**1111**').

Both **A000** and **FFFFA000** are decimal **-24,576**.

The **fullword 0000A000** is **decimal 40,960**, *so not the same as the **halfword A000.***

The correct answer is 'False'.

Question **5**

Incorrect

0.00 points out of 1.00

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The decimal number **-**8,000 can be represented as (pick the two correct answers)

Select one or more:

- [ ] A. Fullword FFFFE0C0
- [x] B. Halfword 1F40 ✖
- [x] C. Fullword 00001F40 ✖
- [ ] D. Doubleword FFFFFFFFFFFFE0C0
- [ ] E. Doubleword 00000000FFFFE0C0

To find the hexadecimal representation of **-**8000, we first convert 8000 to hexadecimal and then change its sign (find its two's complement):

Decimal 8000 to hexadecimal:

8000 / 16 = 500 , with remainder **0**
500 / 16 = 31 , with remainder **4**
31 / 16 = 1 , with remainder 15 (hex **F**)
1 / 16 = 0 , with remainder **1**

Decimal 8000 is hexadecimal **1F40**. Converting 1F40 to negative:

Positive:	1F40	0001 1111 0100 0000
Invert bits:	E0BF	1110 0000 1011 1111
Add 1:	**E0C0**	**1**110 0000 1100 0000 (note the sign)

To represent a **fullword**, we propagate the sign: **FFFFE0C0**

Same for a **doubleword**: **FFFFFFFFFFFFE0C0**

The correct answers are: Doubleword FFFFFFFFFFFFE0C0, Fullword FFFFE0C0